Ruthwik Reddy Sunketa

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Summary

Computer Engineering PhD student specializing in FPGA architecture, design, and automated EDA flows. Current research focuses on next-generation FPGA architectures for AI/ML workload acceleration and hardware-software co-design. Experience in digital design, verification, physical design, and EDA flow development.

Education

Arizona State University — Tempe, AZ

- **Ph.D.** in Computer Engineering — Aug 2025 – Present
- **Master's** in Computer Engineering — Aug 2023 – July 2025

Relevant Coursework: Reconfigurable Computing, VLSI Design, Hardware Security and Trust, Advanced Hardware and Systems for Machine Learning, Statistical Machine Learning, Real-Time Digital Signal Processing

Academic Research Experience

HLS-to-VTR: Architecture-Aware High-Level Synthesis — Dec 2025 – Present

- Developing HLS frontend for VTR framework to enable architecture-aware RTL generation from C/C++ designs
- Implementing custom MLIR lowering passes to automate the mapping of high-level algorithmic operations to specialized hardware like tensor slices and Processing-In-Memory (PIM) blocks

Azure-Lily: Analog In-Memory Computing (IMC) FPGA Architecture — Aug 2025 – Nov 2025

- Conducted design space exploration to integrate RRAM-based IMC engines into a custom FPGA fabric, defining optimal block dimensions, I/O density, and routing interfaces
- Developed Verilog benchmarks for LeNet, ResNet, VGG, and Transformer Attention to evaluate AI workload performance on the hybrid architecture
- Achieved a 6.58× latency reduction and 8,741× energy efficiency gain over baseline FPGA for DNN inference

Configurable Voltage FPGA — Aug 2025 – Jan 2026

- Explored mixed-voltage FPGA architectures with on-chip Digital LDOs (DLDOs) to enable drop-in replacement for legacy 3.3 V / 5 V ICs
- Designed and evaluated a custom FPGA fabric using OpenFPGA, including architecture selection, scalable fabric generation, and bitstream-based functional verification
- Performed RTL-to-GDS physical design using OpenLane, including power analysis, DLDO sizing, DRC/LVS sign-off, and tapeout submission on SKY130 technology

OpenFPGA-NoC: Automated Fabric and Bitstream Generation for NoC-based FPGAs — Aug 2024 - June 2025

- Extended OpenFPGA framework to support hardened Network-on-Chips (NoCs) with arbitrary physical topologies
- Developed a dedicated NoC bitstream generator and configuration mechanism to manage dynamic compile-time address translation and global routing mode selection
- Evaluated NoC-based architectures through an automated RTL-to-bitstream flow, confirming a 3.66× frequency gain and 15% lower routing congestion for I/O-intensive workloads over traditional fabrics

Publications

[1] **Ruthwik Reddy Sunketa**, Muhammad Ali Farooq, Ganesh Gore, Allen Boston, Pierre-Emmanuel Gaillardon, and Aman Arora, ***OpenFPGA-NoC: Automated Fabric and Bitstream Generation for NoC-based FPGAs***, ACM TRETS, 2025. [Journal]

[2] **Ruthwik Reddy Sunketa** and Aman Arora, ***Closing the Loop on FPGA Verification: An Iterative Framework for Maximizing Routing Resource Coverage***, ACM ISFPGA, 2026. [Poster]

[3] Archit Gajjar, **Ruthwik Reddy Sunketa**, Lei Zhao, Omar Eldash, Aishwarya Natarajan, Giacomo Pedretti, Aman Arora, Paolo Faraboschi, Jim Ignowski, and Luca Buonanno, ***Azure-Lily: An FPGA Architecture with Analog IMC Engines for Efficient AI***, ACM TACO, 2026. [Journal]

Technical Skills

Programming: Python, C, C++, Verilog, SystemVerilog, Embedded C

Technologies: Quartus & Xilinx FPGA tools, Synopsys tools (VCS, Design Compiler, PrimeTime), Cadence tools (Genus, Innovus, Virtuoso), Open-source EDA tools (VTR/VPR, OpenFPGA, OpenROAD/OpenLane, Yosys)